EXP 5

CODE:

```
from collections import deque


def valid(mL, cL, mR, cR):
  # Check boundaries
  if mL < 0 or cL < 0 or mR < 0 or cR < 0: return False
  if mL > 3 or cL > 3 or mR > 3 or cR > 3: return False
  # Missionaries must not be outnumbered
  if (mL > 0 and mL < cL) or (mR > 0 and mR < cR): return False
  return True


def next_states(state):
  mL, cL, mR, cR, boat = state
  moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]
  for m,c in moves:
    if boat == 'L':
      ns = (mL-m, cL-c, mR+m, cR+c, 'R')
    else:
      ns = (mL+m, cL+c, mR-m, cR-c, 'L')
    if valid(*ns[:-1]):
      yield ns


def solve():
  start, goal = (3,3,0,0,'L'), (0,0,3,3,'R')
  q = deque([[start]]); seen = {start}
  while q:
    path = q.popleft()
```

```
    if path[-1] == goal: return path
    for s in next_states(path[-1]):
      if s not in seen:
        seen.add(s); q.append(path+[s])

# Run and print solution
for step in solve():
  print(step)
```

OUTPUT:

```
============ RESTART: C:\Users\arunp\OneDrive\Documents\AI\EXP 5.py =========
Initial State: ('A', 'dirty', 'dirty')
Steps to goal:
- Clean room A
- Move to room B
- Clean room B
```